Progress Report: Smart Veterinary System with AI-Based Disease Detection

Course Code: CSE 299
Section: 04
Project Group: 05

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***Abstract*—This progress report presents the current development status of the Smart Veterinary System with AI-Based Disease Detection. The project aims to develop a web-based platform that predicts possible pet diseases using symptom-based text input and image analysis. The system is being implemented using the Django framework and machine learning techniques. This report outlines the technical approach, system architecture, team responsibilities, and completed milestones.**

I. Current Progress

The following tasks have been completed today, focusing specifically on the **HTML pages (frontend) and **backend handling** for user management (login, registration, logout):**

TABLE I
Today's Progress and Responsibilities

Completed Task	Responsible Member(s)
Designed and developed the homepage layout and structure (HTML, CSS)	Rishat Bin Sultan
Created the registration page with form validation and user input handling (HTML, CSS)	Tamim Hasan
Developed the login page with backend authentication handling (Django views, templates)	Fariha Islam
Implemented user authentication backend (login, registration, logout) using Django	Fariha Islam
Set up the database models for user authentication (Django models)	Fariha Islam
Configured Django views to handle user registration and login requests	Tamim Hasan
Implemented user session management and logout functionality	Rishat Bin Sultan

II. Challenges and Initial Decisions

Challenges:

- **Ensuring smooth integration between frontend and backend components, particularly the login and registration features.**
- **Addressing issues with user input validation and session management for secure user authentication.**

Initial Decisions:

- **We will continue refining the frontend and backend integration by focusing on the responsiveness and error handling in user forms.**
- **The backend session handling will be further tested to ensure secure login/logout functionality.**

III. Future Plan

In the upcoming phase, we will:

- **Implement the AI-based disease prediction module and integrate it into the system.**
- **Continue improving the frontend UI and add the dashboard section for user interaction.**
- **Start integrating the AI model with the backend for disease prediction based on user inputs.**
- **Perform end-to-end testing of the login and registration system.**
- **Begin implementation of the online shop module, including product catalog, cart functionality, and checkout process.**
- **Integrate the shop module with the user authentication system for order tracking and purchase history.**

IV. Conclusion

Today's work focused on frontend development (HTML pages for login, registration, and homepage) and backend integration for user authentication. Significant progress has been made in user management functionalities, and the next phase will focus on integrating the Shop and Ai module and further refining system integration.